

KAVIYA A

Electronics & Communication Engineer | Software Development • Embedded Systems • Digital Design

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering 2026 graduate with strong programming fundamentals in C, Java, and Python, and hands-on experience across software development, embedded systems, and digital design. Skilled in analytical problem-solving, clear communication, and collaborative teamwork, with a strong willingness to learn and adapt across domains.

TECHNICAL SKILLS

Programming Languages: C, Java, Python

Software Development: Data Structures & Algorithms, Problem-Solving

Embedded Systems & VLSI: Embedded C, Microcontrollers, FreeRTOS, Communication Protocols, Verilog HDL

Systems & Tools: Linux (Ubuntu), MATLAB, Xilinx Vivado, VS Code

Core Strengths: Analytical Thinking, Communication & Interpersonal Skills, Collaborative Teamwork, Adaptability

TECHNICAL EXPERIENCE

Anti-Jamming Wireless Communication using Star-to-Mesh Topology

2025 – 2026

Skills: Linux(Ubuntu), NS-3 Simulation, netAnim

- Developed a self-healing star-to-mesh network topology to mitigate jamming attacks in wireless communication, improving signal stability across nodes.
- Applied analytical problem-solving to design and validate the system, ensuring reliable data transmission under simulated jamming conditions.

AI-Powered Delivery Slot Customization

2024 – 2025

Skills: Python, Flask, GenAI

- Built a full-stack Python application for a doorstep delivery service, enabling customizable, time-bound delivery slots that reduced missed deliveries and resource wastage.
- Integrated AI-driven real-time customer updates, improving delivery tracking and overall customer satisfaction.

IoT Smart Sewage Gas Monitoring & Alert System

2023 – 2024

Skills: Embedded C, Gas Sensors, MQTT, Cloud Alerting

- Built an IoT-integrated system using CO, CH4, and H2S gas sensors to detect and monitor toxic gas levels inside sewage infrastructure.
- Implemented MQTT-based cloud alerting to notify gas level variations in real time, helping ensure human safety.

EDUCATION

B.E., Electronics & Communication Engineering

2022 – 2026

Sri Ramakrishna Engineering College, Coimbatore — CGPA: 7.88

Relevant Coursework: Programming (C, Java, Python, DSA), Digital Systems, Computer Architecture, Operating Systems, Embedded Systems, Communication Protocols

CERTIFICATIONS

- OCI Certified AI Foundations Associate — Oracle Cloud Infrastructure
- UiPath Academy Specialized AI Associate Training
- MongoDB — 11 Skill Badges in Database Management

EXTRACURRICULAR & LEADERSHIP

- NSS Volunteer, SREC — Community service leadership, structured planning, cross-functional coordination.
- Member, Q-Zenith Quiz Club — Analytical thinking under pressure.

LANGUAGES

English — Fluent (Professional) | Tamil — Native